ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** zero	cos 90 = 0.
2	**(a)** 9	0.06 x 150.
3	**(d)** 4	Effort arm/load arm.
4	**(c)** 20 m/s	sqrt(2gh).
5	**(b)** 120 J	Only vertical displacement counts.
6	**(a)** 45 rupees	9 x 5.
7	**(c)** 50 N	L/MA.
8	**(c)** 36 per cent	(5 - 3.2)/5.
9	**(d)** 5 m/s	v = sqrt(2KE/m).
10	**(a)** 500 W	15000/30.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	W = 200 J (1); KE = 200 J (1).	1+1
12	KE = 450 J (1); F = 450/0.03 = 15000 N (1).	1+1
13	MA = 4 (1); Effort = 100 N (1).	1+1
14	W = 500 J (1); P = 100 W (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Start: KE 225 J (1); At 5 m: PE 100 J, KE 125 J (1); Top: PE 225 J (11.25 m) (1).	1+1+1
16	At 50 cm: 40 N, MA 2.5 (1); At 1 m: 20 N, MA 5 (1); Longer effort arm gives larger MA (1).	1+1+1
17	KE is lost on impact (1); Converts to heat, sound, deformation (1); Energy is conserved overall (1).	1+1+1
	OR: W = 100000 J (1); P = 2500 W (1); mgh (1).	1+1+1
18	Work in = work out (1); E x 5 = 300 x 1 (1); E = 60 N; MA = 5 (1).	1+1+1
	OR: PE lost = 600 J (1); KE gained = 360 J (1); Work against friction = 240 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Start: PE 900 J (1); After 1 s: fallen 5 m, KE 100 J, PE 800 J (1); After 2 s: fallen 20 m, KE 400 J, PE 500 J (1); Ground: KE 900 J (v = 30 m/s) (1); Total 900 J (1).	1+1+1+1+1
	OR: MA = 3 (1); Efficiency = 75 per cent (1); Output work = 600 J (1); Input = 800 J (effort moves 8 m) (1); Loss = 200 J (1).	1+1+1+1+1
20	PE of water to KE to electricity (1); Power = 5000 x 10 x 40 = 2 x 10^6 W (1); Useful power = 1.6 x 10^6 W (1); Efficiency 80 per cent (1); Renewable (1).	1+1+1+1+1
	OR: Work A = 240000 J, P = 400 W (1); Work B = 200000 J, P = 555.6 W (1); B is more powerful (1); Ratio A:B = 0.72 (1); P = mgh/t (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	L/E.
2	**(a)** 2 m/s	sqrt(2 x 10 x 0.2).
3	**(c)** four times	KE proportional to v squared.
4	**(b)** 200 W	W/t.
5	**(a)** 4	l/h.
6	**(b)** 45 m	u squared/2g.
7	**(b)** half	2 x 1/4.
8	**(c)** 1492 W	2 x 746.
9	**(c)** 80 per cent	MA/VR.
10	**(d)** elastic potential energy	Stored.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Larger KE (1); Work = F s = KE, so s larger (1).	1+1
12	Friction and weight of moving parts waste energy (1); Output work is less than input (1).	1+1
13	v = sqrt(2 x 10 x 5) (1); = 10 m/s (1).	1+1
14	Gain in PE is the same (1); Work in = mgh in both cases (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	PE = KE (1); v = 3 m/s (1); PE to KE (1).	1+1+1
16	v = 10 m/s (1); KE = 150 J (1); Fall in 1 s: 5 m, PE lost 150 J (1).	1+1+1
17	Some KE is lost to heat and sound (1); Less energy for the next rise (1); Total energy is conserved (1).	1+1+1
	OR: KE1 = 1000 J (1); KE2 = 4000 J (1); Work = 3000 J (1).	1+1+1
18	Friction: negative work (1); Centripetal force perpendicular to displacement: zero work (1); Moon in circular orbit: zero (1).	1+1+1
	OR: 12 J (1); 12 J (1); Total 24 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	W by man = 1500 J (1); Against gravity = 1000 J (1); Against friction = 500 J (1); Efficiency = 66.7 per cent (1); MA = 500/150 = 3.3 (ideal 5) (1).	1+1+1+1+1
	OR: mgh = (1/2) m v squared (1); v = sqrt(2gh) (1); h = 5: 10 m/s (1); h = 20: 20 m/s (1); Independent of mass and shape (1).	1+1+1+1+1
20	Moment of load = moment of effort (1); MA = effort arm/load arm (1); Effort arm = 1.3 m (1); Effort = 600 x 0.2/1.3 = 92.3 N (1); MA = 6.5 (1).	1+1+1+1+1
	OR: KE = 625 J (1); F = 6250 N (1); t = 0.0008 s (1); At 500 m/s: KE 2500 J, F 25000 N (4 times) (1); Distance four times if the same force (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0	v = 0.
2	**(b)** 14.4 x 10^6 J	2000 x 7200.
3	**(c)** 80 per cent	Output 200 J, input 250 J.
4	**(c)** 500000 J	mgh.
5	**(a)** 80 J	mg x 4.
6	**(d)** twice the power of B	Same work, half the time.
7	**(a)** 100 per cent	No losses.
8	**(a)** 17.3 m/s	sqrt(2 x 10 x 15).
9	**(a)** 8.9 m/s	sqrt(2 x 10 x 4).
10	**(b)** 5000 W	P = F v.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Output 800 J, input 1000 J (1); Efficiency 80 per cent (1).	1+1
12	0.5 kWh (1); 1.8 x 10^6 J (1).	1+1
13	By man 100 J (1); By friction -40 J (1).	1+1
14	Longer effort arm (1); Less effort needed (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Saved power 32 W (1); 32 x 5 x 30 = 4.8 kWh (1); Saving = 28.8 rupees (1).	1+1+1
16	W = 60000 J (1); P = 100 W (1); = 0.13 hp (1).	1+1+1
17	Moments equal (1); 30 x 2 = 40 x d (1); d = 1.5 m (1).	1+1+1
	OR: Hammer has KE at the moment of impact (1); KE gives large force over short distance (1); Work done on the nail = F x d (1).	1+1+1
18	E = 2 x 0.5 x 30 = 30 kWh (1); Cost = 180 rupees (1); Units = kWh (1).	1+1+1
	OR: MA = R/r = 4 (1); Effort = 50 N (1); Ideal machine (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At the top PE max (1); At the valley v = sqrt(600) = 24.5 m/s (1); At the 20 m hill: v = sqrt(2 x 10 x 10) = 14.1 m/s (1); KE and PE exchange (1); Friction ignored (1).	1+1+1+1+1
	OR: At the top: gravitational PE max (1); Falling: PE to KE (1); Trampoline: KE to elastic PE (1); Then back to KE and PE (1); Some energy lost as heat/sound (1).	1+1+1+1+1
20	F x 10 = 5 x 10 x 4 (1); F = 20 N (1); W = 200 J (1); PE = 200 J (1); MA = 2.5 (1).	1+1+1+1+1
	OR: Work = 100000 J (1); Human: 1000 s (1); Motor: 100 s (1); Engine: 1 s (1); Time = W/P (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.